

Instructions to the Students:

- 1. All the questions are compulsory.*
 - 2. Use of non-programmable scientific calculators is allowed.*
 - 3. Assume suitable data wherever necessary and mention it clearly.*
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Q. 1 Solve Any Two of the following. (6 x 2) 12M

- A) Explain Otto cycle with the help of P-V & T-S diagram; Also derive an expression for its thermal efficiency.
- B) Describe the stages of combustion in SI engines
- C) With neat sketch explain Battery Ignition system. Also write the functions of each component.

Q.2 Solve Any Two of the following. (6 x 2) 12M

- A) Explain the construction and working of Four strokes CI engine with neat sketch.
- B) Enlist the different components of IC engine. Also differentiate between 4-stroke and 2-Stroke engine.
- C) Describe various Pollutants from SI & CI engines and their control.

Q. 3 Solve Any Two of the following. (6 x 2) 12M

- A) Draw and explain ideal vapor compression refrigeration cycle on T-S and P-h diagram.
- B) Explain ideal Rankine cycle on P-V, T-S & h-s diagram also derive the expression for thermal efficiency of cycle.
- C) Differentiate between horizontal axis wind turbine and vertical axis wind turbine.

Q.4 Solve Any Two of the following. (6 x 2) 12M

- A) Describe briefly with the help of a diagram, the vapor absorption refrigeration system.
- B) Explain the terms in details:
 1. Dry Bulb Temperature
 2. Dew point temperature
 3. Relative Humidity
- C) Explain steam power plant with the help of schematic diagram. Also write down the functions of each component.

Q. 5 Solve Any Two of the following.

(6 x 2) 12M

- A) Describe the diesel power plant with the help of schematic diagram.
- B) Draw general layout of Hydroelectric power plant. Also explain functions of its component.
- C) In a S.I Engine working on the ideal Otto cycle, the compression ratio is 5.5. The pressure and temperature at the beginning of compression are 1 bar and 27°C respectively. The peak pressure is 30 bar. Determine the pressure and temperatures at the salient points, the air-standard efficiency and the mean effective pressure. Assume ratio of specific heats to be 1.4 for air.

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